



Toolboxes for creating immersive virtual reality experiments in psychological, behavioral, and cognitive research: A scoping review

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ABSTRACT

Background: Immersive VR, especially head-mounted displays, allows ecologically representative research in human psychology, behavioral sciences, and cognitive neuroscience. Custom VR environments provide considerable experimental control, but their development is often resource-intensive. Virtual reality research is now more accessible thanks to software development kits and platforms. Despite their growing use, these toolboxes lack a systematic overview of their usability and application, hindering informed decision-making by researchers.

Objective: The current scoping review maps the landscape of existing VR toolboxes for immersive cognitive, behavioral, and psychological research. Unlike previous reviews that focus on specific experimental niches, this study provides a comprehensive evidence map and a practical decision support framework specifically tailored for non-programmer researchers.

Methods: In December 2024, we searched scientific databases and grey literature using PRISMA-ScR principles. We found peer-reviewed English-language papers on immersive VR authoring tools for human behavioral research using a preset search strategy and a multi-reviewer method to assure reliable screening and data extraction. We mapped toolbox features, major research applications, and non-programmer researcher usability using descriptive and thematic analysis of retrieved data.

Results and discussion: Our synthesis identified 34 different toolboxes, revealing that the most common applications are for general purposes, spatial navigation, and social cognition. This review provides a decision matrix that categorizes toolboxes by technical requirement, research compatibility, and documentation quality. These insights can assist developers and researchers in addressing challenges such as complex scripting requirements and inconsistent documentation. The thorough evidence map and a research agenda aimed at improving the development and application of effective, user-friendly virtual reality tools within the behavioral and brain sciences.

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1. Introduction

1.1. The increasing significance of VR in behavioral research

Virtual reality (VR) is attracting growing interest due to its potential to provide immersive experiences across fields like entertainment, health, and education (Freeman et al., 2017; Radianti et al., 2020; Slater & Sanchez-Vives, 2016). Its use is particularly instrumental in the behavioral and psychological fields, serving as a transformative medium for studying brain health and treating disorders like depression and PTSD (Emmelkamp & Meyerbröker, 2021; Montesano & Seinfeld, 2025; Rizzo et al., 2013). A strong body of literature supports its use by neuroscientists to investigate and design experiments for different cognitive and executive functions in dynamic environments (Basdogan et al., 2025; Kozlov & Johansen, 2010; Mertens & van Gelder, 2025).

By simulating naturalistic environments with high ecological validity, VR gives users a powerful sense of “presence” that transforms them from passive observers into active ones (Parsons, 2015). Its high level of interactivity, which enables safe, repeatable, and cost-effective experimentation, makes it an ideal tool for studying behavior under controlled conditions (de la Rosa & Breidt, 2018). Additionally, VR offers the unique opportunity to immerse users in scenarios that would be impractical or unethical to replicate in the real world (Francis et al., 2017).

This capability marks a substantial shift from conventional research methods. For many years, static, two-dimensional stimuli, like word lists on a screen, pairs of images, or basic objects on a computer screen, were the mainstay of cognitive research. While foundational, this approach is essentially limited, as it does not capture how cognition functions in dynamic, multisensory environments (Parsons, 2015; Tarr & Warren, 2002). By immersing participants *inside* a controlled and interactive world where the “scene” itself serves as the independent variable, virtual reality gets around this issue. Immersive VR allows for the study of complex behavior in contexts that would be impractical or unethical to recreate in real life, such as high-risk decision-making or therapeutic interventions for PTSD (Blasovich et al., 2002; Bohil et al., 2011). As evidenced by a rapid rise in VR-related publications since 2013, virtual reality (VR) is rapidly becoming a substantial tool and field of research in human behavior (Cipresso et al., 2018).

1.2. The challenge: the technical obstacle to VR experimentation

Despite its transformative potential, many researchers and research labs are unable to utilize virtual reality due to significant technical barrier limits (Alsbury-Nealy et al., 2022). While powerful game engines like Unity provide a range of tools, assets, and add-ons to help create interactive worlds, these resources are typically designed for commercial or entertainment purposes and are not optimized for scientific research. They frequently lack the essential and built-in features required for rigorous research, such as precise temporal control, sophisticated data logging capabilities, and structured trial management (Brookes et al., 2020). As a result, researchers are forced to either attempt to adapt general-purpose tools for a scientific setting or build these essential components from scratch, which requires advanced programming skills, expertise in 3D modeling, and a substantial time and resource commitment; this technical barrier often limits the use of VR to large, well-funded labs with specialized technical teams, leaving it inaccessible to smaller research groups or interdisciplinary scholars who might not have the necessary software development experience. This poses a paradox: the researchers who have the best chance to ask critical questions are often the least equipped with the tools needed to answer them.

1.3. The solution: the rise of specialized VR toolboxes

A rapidly growing ecosystem of specialized VR platforms and

toolboxes has been developed to close this accessibility gap and to simplify the creation of virtual environments for research. By providing accessible interfaces and pre-made modules for interaction, navigation, and data logging, these tools aim to democratize immersive research. Such toolboxes, which include domain-specific toolkits like OpenMaze (Alsbury-Nealy et al., 2022), which simplifies the creation of complex spatial navigation tasks; foundational frameworks that provide a core software structure for building experiments (such as Unity Experiment Framework (UXF)) (Brookes et al., 2020); standalone applications like NeuroVirtual 3D (Pietro et al., 2014), which require no programming; and a specialized stimulus library (Okada et al., 2024), lower technical barriers and allow researchers to design and conduct sophisticated studies without requiring extensive programming knowledge.

1.4. The gap in the literature and the rationale for the review

However, despite their potential and the rapid proliferation of these tools, a critical knowledge gap exists. Researchers have a hard time finding out what tools are available, comparing their technical features, or figuring out if they are right for a certain research area because there is no comprehensive map of the options regarding the strengths, weaknesses, and usability of these toolboxes. This forces researchers to make decisions based on incomplete information, potentially hindering the adoption of VR and leading to duplicated development efforts and the inefficient use of research funding.

1.5. The present study: objectives and research question

This scoping review seeks to identify, characterize, and delineate the landscape of existing specialized VR toolboxes designed for immersive behavioral, psychological, and cognitive research. Guided by the PCC (Population, Concept, Context) framework; where the Population comprises specialized VR software tools for behavioral research, the Concept encompasses their features, capabilities, and adoption patterns, and the Context is immersive VR-based experimentation in behavioral, cognitive, and neuroscientific research—this review addresses the following research questions:

1. What specialized software tools and platforms currently exist for conducting immersive VR-based behavioral, cognitive, and neuroscientific research, and what are their core technical features and functional capabilities?
2. What are the key trade-offs, barriers to adoption, and sustainability challenges associated with these tools, particularly regarding technical expertise requirements, experimental control, and open-source viability?
3. How can existing tools be systematically categorized to guide researchers in selecting an optimal platform based on their specific research domain, team expertise, hardware needs, and project resources?

While our primary focus is on behavioral, psychological, and cognitive research, we recognize the significant methodological overlap between these fields and Human-Computer Interaction (HCI). Many toolboxes included in this review were originally developed for UX testing or general XR experimentation but have been included here because they offer essential capabilities for behavioral science, such as high-fidelity data logging, standardized questionnaire integration, and synchronized physiological event marking.

The review will map these toolboxes along a number of key axes, such as their general features, technical capabilities, data handling protocols, usability, and scientific applications. Our unique contribution lies in the synthesis of technical requirements and functional features into a navigable decision matrix, allowing researchers to select tools that align with both their experimental needs and their team's technical proficiency—an aspect often overlooked in purely technical or

pedagogical reviews. This review aims to accelerate VR adoption, encourage informed tool selection, and push the field's methodological frontier forward.

2. Method

This scoping review was performed and reported in alignment with the PRISMA-ScR (Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews) guidelines (Tricco et al., 2018). The research protocol has been officially registered on the Open Science Framework (OSF, DOI: <https://doi.org/10.17605/OSF.IO/N6JFC>) to ensure transparency, reproducibility, and compliance with open science standards. The registration has a detailed description of the study design, methods, and analysis plan. (Hosseinzadeh et al., 2025, August 9; August 9) The review systematically mapped and assessed the various Virtual Reality (VR) toolboxes employed in the creation of immersive experiments within behavioral, psychological, and cognitive research. Its main goal was to deliver a comprehensive overview of the tools that are available to help design and implementation of studies use head-mounted displays (HMDs).

2.1. Information sources and search strategy

A systematic search was performed through four scientific databases, including IEEE Xplore, PubMed, Scopus, and Web of Science in December 2024. To identify reports not indexed in these databases, supplementary grey literature searches were conducted on preprint servers (e.g., arXiv, PsyArXiv) and Google Scholar. These supplementary searches were not systematic; rather, they employed targeted keyword searches to identify any additional relevant tools or frameworks that may have been missed by the primary database search. Screening was performed on an ad hoc basis, with records reviewed until no relevant tools were being identified.

In addition to the database and grey literature searches, a citation searching method (snowballing) was employed. This process employed both the ascendancy method (examining the reference lists of included articles) and the descendancy method (analyzing articles that have cited the included articles) to uncover more relevant literature. To ensure the retrieval of relevant studies during the search process, alongside the term “virtual reality,” an exhaustive list of search terms was formulated, necessitating the inclusion of at least one term—such as psychology, cognition, behavioral neuroscience, toolkit, toolbox, framework, or platform—to appear in the title or abstract of the studies obtained. The search query combined keywords related to the technology, research domain, and tool type using Boolean operators. To be included, a publication's title or abstract was required to contain terms related to virtual reality (e.g., “Virtual reality,” VR, HMD), a relevant research field (e.g., psycholog*, cogniti*, behavi*), and the nature of the tool (e.g., toolkit, toolbox, framework). The full query strings used for all databases are shown in [Appendix 1](#).

2.2. Eligibility criteria

Sources were included in the review if they described a software tool—defined broadly as a framework, toolbox, standalone application, plugin, or library—that was either explicitly designed for or had been demonstrably used for creating and conducting experiments to study human behavior, cognition, perception, or motor control in a virtual environment.

Publications were required to meet the following inclusion criteria:

1. **Focus:** The study must describe the development or evaluation of an authoring tool or software designed to create immersive VR experiences for behavioral, cognitive, and psychological experimental research

2. **Technology:** Pertained to immersive VR delivered via an HMD (desktop-based VR or CAVE/IRE projection systems was excluded).
3. **Population:** The tool was intended for, or had been used in research with human participants. Studies focusing solely on animal studies were excluded.
4. **Publication Type:** Inclusion was limited to original research articles published in peer-reviewed journals or conference proceedings. Review articles, book chapters, and editorials were excluded to ensure data was extracted from primary software descriptions.
5. **Language:** Was written in English.

Beyond these categorical filters, a functional justification was applied to account for the multidisciplinary nature of VR software. Tools were included if they provided the necessary technical infrastructure for empirical human research, even if their primary framing in the literature was within Human-Computer Interaction (HCI) or User Experience (UX) contexts. These platforms were deemed essential to the review if they offered frameworks for:

1. Recording participant behavior (e.g., precise data logging and telemetry).
2. Managing experimental trials (e.g., randomization and stimulus control).
3. Assessing subjective states (e.g., integrated in-VR questionnaires).

Excluding these “UX-centric” toolboxes would have omitted some of the most robust and accessible platforms currently utilized by psychological researchers to build their experimental pipelines.

Exclusion criteria were applied to maintain the review's focus on tools for experimental research. Tools were excluded if their primary purpose was: (1) solely for the delivery of clinical therapy or training without a stated component for research or assessment; (2) low-level hardware-specific software development kits (SDKs) that were not part of a larger, integrated research framework or are general purpose VR authoring tool; or (3) tools that were no longer available, accessible, or supported; (4) Studies that described a tool that was a duplicate or a redundant version of an already included toolbox.

2.3. Screening

The study selection process was managed using the Rayyan web application. After an initial automated deduplication of records using EndNote, three independent reviewers performed the screening in two phases. First, the reviewers screened the titles and abstracts of all unique records. To ensure consistency, a pilot screening of 10% of the results was conducted to calibrate decisions. Disagreements were resolved through discussion; if a consensus could not be reached, a senior reviewer made the final decision. In the second phase, the full texts of all articles that passed the initial screening were retrieved and assessed against the eligibility criteria by two independent reviewers, following the same procedure for resolving disagreements. The reasons for excluding articles at this stage were documented.

2.4. Data charting and synthesis

A standardized data extraction form was created using Microsoft Excel to systematically record pertinent information from the final set of included studies. Two reviewers independently extracted data from a 10% subset of articles to pilot the form and ensure a shared understanding of the variables. After this calibration, the reviewers worked together to get data from all the other studies, if there were any disagreements, they were worked out through discussion or by a third reviewer.

The extracted entities directly addressed the predefined primary and secondary research questions. (Hosseinzadeh et al., 2025, August 9; August 9) They covered the general characteristics, technical features,

usability, and research applications of the toolboxes, provided a comprehensive dataset to map the field (see [Appendix 2](#) for the full list of extracted entities). A thematic analysis method was used to classify descriptive data, including reported advantages, limitations. Common themes were identified and grouped to provide a structured overview. The term for technical features (e.g., input devices, supported engines, data export formats) was standardized for frequency counts and thematic grouping. This dual approach of a numerical summary and thematic analysis was chosen to provide both a broad, quantitative picture of the field and a deeper, qualitative understanding of important issues, in line with the methodological framework for scoping reviews.

3. Results

3.1. Search and selection process

The study selection process is summarized in the PRISMA-ScR flow diagram ([Fig. 1](#)). The preliminary search of electronic databases yielded 3686 records. An additional 16 records were identified through grey literature searching. Duplicates and other ineligible records were removed before screening, then 1005 records were screened by title and abstract. This led to the retrieval of 76 full-text reports for eligibility assessment. Following the full-text review, 42 reports were excluded based on following criteria: 7 were in wrong field (Reason 1), 18 did not describe a toolkit (Reason 2), 11 focused on non-headset systems like CAVE/IRE - Cave Automatic Virtual Environments or Immersive Room Environments (reason 3), and 6 were describing same tools as those previously has been included. Finally, a total of 34 reports was included in this scoping review, of which 22 reports were from database searching and 12 reports from other searching methods.

3.2. General characteristics of identified VR toolboxes

The studies and toolboxes covered in this review introduced 34 different VR toolboxes and platforms designed for immersive behavioral, psychological, or cognitive research. [Table 1](#) gives a general picture of the VR toolboxes that were found, detailing their publication timeline, underlying technology, licensing, availability, and support structures.

3.2.1. Publication timeline and geographic distribution

The development of these toolboxes started from 2009 and include all the way till 2024 (though foundational platforms like WorldViz/Vizard may date back to as early as 2002 ([WorldViz, 2023](#))). However,

the process has sped up a lot in recent years. A significant majority of the toolboxes (25 out of 34, or 73%) were released or received major updates in 2020 or later, this shows how quickly interests of scientific community is growing in creating accessible VR research tools.

Geographically, the development of these toolboxes is concentrated in North America and Europe. Research labs and companies in the United States (9/34, 26%) and Germany and Canada (each 5/34, 15%) are the most frequent contributors, followed by the UK and Italy (each 3/34, 9%). The remaining toolboxes originate from a range of other countries, including France, China, Estonia, Ireland, Czech Republic, and Denmark.

3.2.2. Technology, licensing, and support structures

The Unity game engine is the most common platform, serving as the foundation for half of the toolboxes (17/34, 50%). This is followed by frameworks built on Python (8/34, 23%, e.g., psychXR, Vizard), which often provide researchers with low-level hardware access. A smaller number of toolboxes use custom engines (4/34, 12% e.g., CyberSession, OMVR), the Unreal Engine, or web-based technologies like WebXR. (3/34, 9%).

The ecosystem is strongly dominated by open-source development. A clear majority (24/34, 71%) are available under open-source licenses, most often the MIT and GPL licenses. Mostly can be found via public repositories like GitHub. On the other hand, only 12% (4/34) are explicitly proprietary or commercial. Documentation is provided for nearly all identified toolboxes, and almost all of them offer some form of community support, most frequently through GitHub issues pages, dedicated forums, or direct developer contact, fostering a collaborative environment for users and developers.

3.3. Technical characteristics

The technical features of the 34 identified VR toolboxes are detailed in [Table 2](#). The capabilities differ widely, mirroring the diverse requirements of behavioral and cognitive research.

3.3.1. Environment and experiment creation

The way to create an environment and set up experiment logic are the key distinction among toolboxes. A majority of the toolboxes (25/34, 74%) use scripting (primarily C# or Python) to define experimental logic. This gives the maximum flexibility at the cost of requiring programming expertise. To lower this technical barrier, 47% of toolboxes (16/34) offer some kind of GUI-based solution. Some are completely no-code environments like NeuroVirtual 3D, which has a drag-and-drop

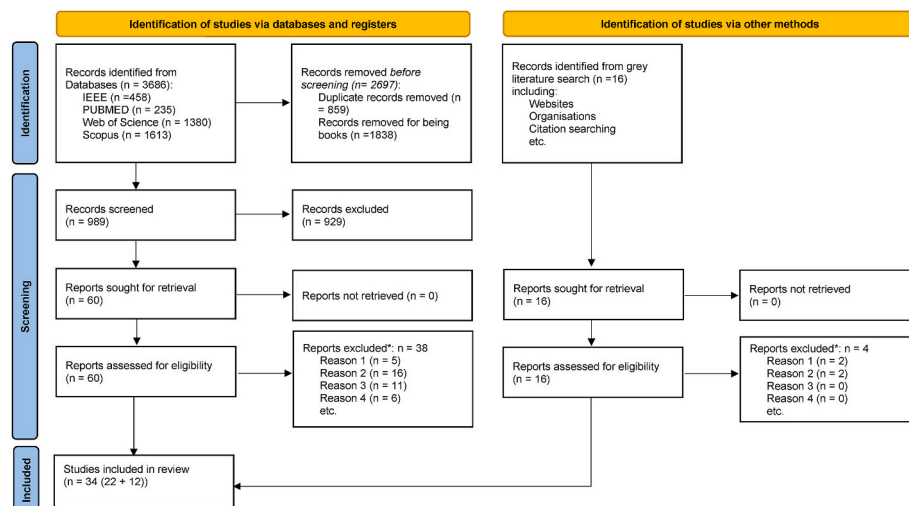


Fig. 1. PRISMA 2020 flow diagram for new systematic reviews which included searches of databases, registers and other sources.

Table 1
Overview and general characteristics of identified VR toolboxes.

Toolbox Name	Reference	Country	Type ^a	Technology	Licensing	Availability and Link	Primary Purpose	DOCUMENTATION AND Community Support
Attention Atlas (AA)	Norwood et al. (2023)	Australia	Standalone Project	Immersive VR, eye-tracking	CC BY-NC-ND	GitHub (https://github.com/davidrosspainted/TheAttentionAtlas) and OSF (https://osf.io/pa96f/)	Visuospatial neglect; Attention mapping; Rehabilitation	Documentation (Guides, Tutorials), Support (via GitHub)
BARGAIN	Benlamine et al. (2021)	Canada	Framework	Unity, EEG, FSM	Not Specified	By request	Affective gaming; Socio-moral development	Documentation (In publication), Support (Not specified).
bmlTUX	Bebko and Troje (2020)	Canada	Framework	Unity, C#	CC BY-NC-SA 4.0	GitHub (biomotionlab.github.io/TUX/)	Vision/perception research; General experiment control	Documentation (Wiki, Tutorials), Support (GitHub, Forums)
BonVision	Lopes et al. (2021)	UK	Plugin	Bonsai, C#, OpenGL	MIT	Bonsai package manager, GitHub (biomotionlab.github.io/TUX/)	Visual neuroscience; Closed-loop experiments	Documentation (Docs, Tutorials), Support (GitHub, Forum)
CyberSession	Not Specified	Not Specified	Standalone App	Custom, Unreal/Source	Proprietary	Commercial, via VTplus	Behavioral research; Exposure therapy; Neuroscience	Documentation (Manuals), Support (Vendor Support)
Doucet_2016	Doucet et al. (2016)	Canada	Framework	UDK, Matlab, Python	Not Specified	By request	Cross-species visual/cognitive experiments	Documentation (In publication), Support (Not specified).
Landmarks	Starrett et al. (2021)	USA	Plugin	Unity, C#	GPL-3.0	GitHub (https://github.com/mjstarrett/Landmarks) and the Human Spatial Cognition Lab website.	3D navigation; Spatial cognition; Memory	Documentation (Wiki, Tutorials), Support (GitHub)
luVRre	Schöne et al. (2021)	Germany	Stimulus Library	3D/360° Video	Open-source	By request	Emotion; Motivation; Cognitive research	Documentation (In publication), Support (Email)
MovementVR	Rossi et al. (2025)	USA	Standalone App	Unity	Open-source	Project website (movementvr.github.io)	Motor control; Motor learning; Reaching	Documentation (Docs, Guides), Support (GitHub)
NavWell	Commings et al. (2020)	Ireland	Standalone App	Windows/MacOS, Web admin interface	Freely	Project website (navwell.cs.nuim.ie/home)	Spatial navigation; Memory; Clinical research	Documentation (Manuals, Videos), Support (email)
NeuroVirtual 3D	Cipresso et al. (2016)	Italy	Platform	NeuroVR 2.0; OpenGL	Open-source	Project website (www.neurovirtual.eu)	Psychometric assessment; Clinical experiments	Documentation (Manuals, Protocols), Support (Project Team)
NeuroVR 2	Riva, G. et al. (2011)	Italy	Platform	C++, Delta3D	Free	Project website (http://www.neurovr2.org)	Behavioral health; Clinical psychology	Documentation (User Guides), Support (Dev Team)
Okada et al. Paradigm	Okada et al. (2024)	USA	Stimulus Library	Unity, Web (Qualtrics)	Open Access (OSF)	OSF (https://osf.io/s68rc/)	Episodic memory; Scene recognition; Déjà vu	Documentation (In publication), Support (OSF)
OMVR	Pedersen, R. L. et al. (2023)	Denmark/UK	Platform	Custom (Oticon)	Proprietary,	Via Oticon Medical, for research partners/ collaborators	Hearing science; Auditory perception	Documentation (In publication), Support (Not specified).
OpenMaze	Alsbury-Nealy et al. (2022)	USA/Canada	Framework	Unity, C#	MIT	GitHub (github.com/DuncanLab/OpenMaze)	Spatial navigation; Memory; Cognitive neuroscience	Documentation (Manual, Tutorials), Support (GitHub, Lab)
Ouvrai	Cesaneck et al. (2024)	USA	Web Platform	JS, WebXR, Three.js	GPL-3.0	Website (ouvrai.com), GitHub (github.com/EvanCesaneck)	Remote behavioral neuroscience; Motor learning	Documentation (Website, Demos), Support (GitHub)
PLUME	Javerliat et al. (2024)	France	Framework	Unity, Python, Web	GPL-3.0	GitHub (https://github.com/liris-xr/PLUME)	Behavioral/physiological data collection (6DoF XR)	Documentation (GitHub), Support (GitHub, Discord)
psychXR	Not Specified	Not Specified	library	Python, OpenGL	MIT	PyPI, GitHub (https://github.com/mdcutone/psychxr)	Vision science; Neuroscience; Low-level hardware access	Documentation (Full Docs, API), Support (GitHub, Forums)
PTVR	Castet et al. (2024)	France	Framework	Python, Unity	GPL-3.0-or-later	Website (ptvr.inria.fr)	Vision science; Psychology; HCI; Clinical research	Documentation (Full Docs, Tutorials), Support (Mailing List)

(continued on next page)

Table 1 (continued)

Toolbox Name	Reference	Country	Type ^a	Technology	Licensing	Availability and Link	Primary Purpose	DOCUMENTATION AND Community Support
RemoteLab	Lee et al. (2022)	USA	Plugin	Unity, C#, Photon	MIT	GitHub (https://github.com/microsoft/Remote-Lab)	Remote, multi-user VR experiments	Documentation (Full Guides), Support (GitHub)
ROVER	Küntzer et al. (2024)	Germany	Toolkit	SteamVR Overlay	CC BY 4.0	GitHub (https://github.com/kuentzel/ROVER)	In-VR questionnaires; User experience research	Documentation (Guides, Samples), Support (GitHub)
SALLO	Esposito et al. (2023)	Italy	Toolbox	Unity, C#	GPL-3.0	GitHub (https://github.com/DavideSpot/SALLO)	Spatial orientation, psychophysics, perception	Documentation (GitHub, Examples), Support (GitHub)
SceneExplorer	Not Specified	Not Specified	Web Platform	Unity/ Unreal, Web	Proprietary	Subscription via Cognitive3D, provides free version for limited or academic use	Post-hoc analysis; User behavior visualization	Documentation (Customer Guides), Support (Vendor, Discord)
SkyBXF	Liu & Chen, 2022	China	Game Mod	Creation Engine	GPL-3.0	GitHub (https://github.com/ZeminL/SkyBXF)	Behavioral research in commercial games	Documentation (Setup Guides), Support (GitHub)
SNaP Framework	Annett and Bischof, 2009	Canada	Framework	Virttools, Python, XML, VRPN	Not specified	Not specified	Spatial navigation; Cognitive neuroscience; Aging	Documentation (In publications), Support (Research Group)
Synapse-Unity API	Montag et al. (2021)	USA	API/Library	Unity, C#, TDT	Permissive	GitHub (Forthcoming)	Closed-loop BCI experiments; Neural signals in VR	Documentation (In publication), Support (Research Group)
Toggle Toolkit	Ugwitz et al. (2021)	Czech Republic	Toolkit	Unity, C#	Open-source	GitHub (https://github.com/kim-dotcom/toggleScript)	Controlling interactive 3D experiments; Behavior tracking	Documentation (Video Tutorial), Support (GitHub)
UXF	Brookes et al. (2020)	UK	Framework	Unity, C#, WebXR	MIT	GitHub (github.com/immersivognition/unity-experiment-framework)	General human behavior experiments in VR	Documentation (Tutorials, Wiki), Support (GitHub)
vexptoolbox	Schuetz et al. (2022)	Germany	Library	Python, Vizard	MIT	GitHub (https://github.com/ischtz/vizard-experiment-toolbox)	Trial-based behavioral experiments	Documentation (Manual, Tutorials), Support (GitHub)
VNT	Müller et al. (2023)	Germany	Toolbox	Unity, C#	GPL-3.0	GitLab (https://gitlab.uni-bielefeld.de/virtual_navigation_tools/unity_vnt_main)	Spatial navigation; VR-based human research	Documentation (Wiki), Support (GitLab)
Vizard	Not Specified	USA	Dev Environment	Custom, Python OpenGL, IDE	Commercial	Website (https://www.worldviz.com/vizard-virtual-reality-software)	Rapid development; Research; Training	Documentation (Full Manuals), Support (Vendor, Forums)
VRAT	Bagherzadeh and Tehrani (2024)	USA	Platform	Custom	Not Specified	GitHub (https://github.com/HCAI-Lab/Virtual-Reality-Analysis-Tool-VRAT-)	Behavioral/ cognitive research; Clinical assessment	Documentation (In publications), Support (Collaboration)
VREX	Vasser et al. (2017)	Estonia	Plugin	Unity, C#	CC BY-SA	Website (vrex.mozello.com)	Experimental psychology; Neuroscience	Documentation (Guides, Videos), Support (Implied)
VRSTK	Wölfel, M. et al. (2021)	Germany	Toolkit	Unity, C#	MIT	GitHub (https://github.com/experience-lab/VRSTK)	Human factors; Human behavior in virtual worlds	Documentation (GitHub), Support (GitHub)

API: Application Programming Interface. **BARGAIN:** behavioral affective rule-based games adaptation interface. **BCI:** Brain-Computer Interface. **bmlTUX:** Bio-motionLab Toolkit for Unity Experiments. **CC BY 4.0:** Creative Commons Attribution 4.0 License. **CC BY-NC-SA 4.0:** Creative Commons Attribution-NonCommercial-ShareAlike 4.0 License. **CC BY-SA:** Creative Commons Attribution-ShareAlike License. **EEG:** Electroencephalography. **FSM:** Finite State Machine. **GPL-3.0:** GNU General Public License v3.0. **HCI:** Human-Computer Interaction. **IDE:** Integrated Development Environment. **JS:** JavaScript. **IuVR:** Library for universal virtual reality experiments. **MIT:** Massachusetts Institute of Technology (referring to the license type). **OMVR:** Oticon Medical Virtual Reality. **OpenGL:** Open Graphics Library. **OSF:** Open Science Framework. **PTVR:** Perception Toolbox for Virtual Reality. **PUN 2:** Photon Unity Networking 2. **PyPI:** Python Package Index. **ROVER:** Rating Overlay for Virtual Environments in Research. **SALLO:** Suite for the Assessment of Low-Level cues on Orientation. **SkyBXF:** SkyrimVR-based behavioral experimental framework. **SNaP Framework:** Spatial Navigation Paradigm. **TDT:** Tucker-Davis Technologies. **UDP:** User Datagram Protocol. **UDK:** Unreal Development Kit. **UXF:** Unity Experiment Framework. **VNT:** Virtual Navigation Toolbox. **VR:** Virtual Reality. **VRAT:** Virtual Reality Analysis Toolkit. **VRSTK:** Virtual Reality Scientific Toolkit. **VRPN:** Virtual Reality Peripheral Network. **XML:** Extensible Markup Language.

^a **Definition used in "Type" Column:** **Framework/Toolbox/Toolkit:** A foundational set of tools for building various experiments. **Plugin/Library/API:** An add-on or library for an existing engine or language (e.g., Unity, Python). **Platform/Standalone App/Dev Environment:** A more complete, often self-contained application for creating or running experiments. **Stimulus Library/Game Mod:** Provides pre-made assets or modifies existing software for research purposes.

Table 2
Summary of VR toolbox features.

Toolbox Name	Experiment Design	Hardware & I/O	Data Handling	Advanced Features
Attention Atlas (AA)	Environment and logic defined entirely via C# scripting.	Supports HMDs, controllers, and eye-trackers with visual, audio, and haptic output. Integrates with SteamVR, TobiiXR, and Python.	Records behavioral, positional, and eye-tracking data with real-time logging. Exports to multiple formats including C# native and NumPy arrays.	Supports avatars/agents. No multi-user or cross-species support reported.
BARGAIN	Uses a GUI and C# scripting. Logic is defined via a rule-based Finite State Machine (FSM) GUI.	Supports EEG, HMDs, joysticks, and physiological sensors. Output is visual, auditory, and adaptive to user state. Integrates with NeuroExpress for EEG.	Records behavioral data and inferred/self-reported emotional states. Affective states are evaluated every 30 s.	Supports Non-Player Character (NPC) avatars. No multi-user or cross-species support.
bmlTUX	Uses C# scripting. Logic is defined via a GUI for factorial designs and C# scripts for runtime logic.	Supports any device compatible with Unity. Output is visual, audio, and provides real-time feedback.	Records dependent variables and participant demographics. Exports design and results to JSON. Timing is limited by Unity's update rate.	Supports a participant entity avatar. No multi-user or cross-species support.
BonVision	Uses the Bonsai GUI and scripting. Logic is defined via the graphical Bonsai language.	Supports HMDs, cameras, and physiological sensors. Output is visual (2D/3D) and auditory. Integrates with extensive hardware via Bonsai.	Records behavioral, physiological, and positional data with timing logs. Reports ~35ms latency at 60Hz.	Supports avatars/agents and has cross-species (human and mouse) support.
CyberSession	Environment and logic are defined via JavaScript scripting and integration with Unreal/Source engines.	Supports HMDs, eye-trackers, and joysticks. Output is visual and can control external devices.	Records behavioral, positional, and physiological data with timestamps. Exports to log files.	Supports avatars via engine integration and multi-user capabilities.
Doucet_2016	Environment created in the UDK Level Editor. Logic is scripted in Matlab or Python via TCP commands.	Supports mouse-emulated devices and joysticks. Output is stereoscopic visual. Can integrate with any external data acquisition framework.	Records player position, rotation, and state with timestamps. Aims for millisecond temporal precision.	Supports a "Pawn" class for user representation and is designed for cross-species use.
Landmarks	Environment created in the Unity Editor GUI. Logic defined via the GUI Inspector or C# scripting.	Supports HMDs, treadmills, and eye-trackers. Output is visual and spatialized audio. Integrates with Tobii Pro and Microsoft Azure.	Records player position, task state, and behavioral variables with timestamps. Exports to TXT and CSV.	Supports player controller "Characters." Uses Azure for cloud data storage.
luVRre	A pre-recorded 360° video and photo database; no environment creation or logic definition.	Supports any HMD capable of 360° video playback. Used in conjunction with external EEG systems.	N/A (stimulus database).	No advanced features reported.
MovementVR	Uses a GUI (MovementVR Builder) for all experiment and logic design, with no coding required.	Supports HMDs with integrated hand tracking (e.g., Meta Quest 2) and Leap Motion sensors. Output is visual and auditory.	Automatically records and exports kinematic data from hands and other task variables. A GUI analyzer converts data to standard CSV format.	Supports a realistic, articulated hand avatar for the user. No multi-user or cross-species support reported.
NavWell	Environment and experiment logic are both designed using a GUI-based administrator console.	Supports joysticks, keyboards, and HMDs. Output is visual (2D or 3D).	Records latency, distance traveled, heat maps, and track plots. Exports to CSV.	The participant's navigation entity acts as the avatar. No other advanced features.
NeuroVirtual 3D	A GUI-based drag-and-drop editor. Logic is defined via visual menus with no programming required.	Supports a wide range of biosensors, HMDs, Kinect, and eye-trackers. Integrates with MATLAB.	Records psychophysiological signals, coordinates, and timestamps with millisecond precision. Exports to ASCII log files.	Supports imported avatars. Allows for online sharing of environments.
NeuroVR 2	A GUI editor for pre-designed scenes. Logic is defined via a GUI "Action & Trigger" system.	Supports standard computer inputs. Output is visual and auditory. Future support planned for eye-tracking and physiological sensors.	Records behavioral data.	No advanced features reported.
Okada et al. Paradigm	A pre-built paradigm with a full stimulus set of 128 scenes (64 pairs). Logic is defined by the study/test procedure.	Supports various immersion levels (2D screen, HMDs). Input is via keyboard/mouse (2D) or VR controllers. Output is visual and auditory.	Records scene recall, familiarity ratings, déjà vu reports, and turn predictions. Data export via Qualtrics (2D) or custom Unity logging (VR).	Designed for potential integration with fMRI and electrophysiology. No explicit avatar or multi-user support.
OMVR	Uses pre-existing Unity scenes and templates. Logic is defined via JSON configuration files and a Python controller.	Supports HMDs with head/eye tracking. Output is visual and spatialized audio. Integrates with LSL and ASIO.	Records head movement, gaze, pupil dilation, and user input with timestamps. Exports to XDF.	No primary focus on avatars. Supports networked computers for multi-user interaction.
OpenMaze	Uses the Unity GUI and imported models. Logic is defined via JSON configuration files, with C# for advanced functions.	Supports keyboards and joysticks, with VR headset support via add-ons. Integrates with VR/fMRI via augmentation.	Records positional data, goal collisions, and keystrokes. Exports to CSV. Goal collisions logged every 20ms.	Supports online testing by downloading applications.
Ouvrai	Environment and logic are defined via JavaScript scripting (Three.js), using a Finite State Machine (FSM).	Supports HMDs, hand tracking, and standard K/M/touch. Integrates with Firebase, Prolific, and data analysis software.	Records kinematic, behavioral, and positional data with timestamps. Exports to JSON and CSV. High timing precision (~8.4ms sample period).	Supports imported animated avatars and has a cloud database via Firebase.
PLUME	A Unity plugin. Logic is defined via C# scripting.	Supports HMDs, eye-trackers, and physiological sensors via Lab Streaming Layer (LSL).	Records behavioral, physiological, and custom event data. Exports to multiple formats (CSV, XDF, etc.). Has high precision (100ns clock accuracy).	Records user motion for avatars and supports multi-user setups.
psychXR	Environment and logic are defined via Python scripting with an OpenGL framework.	Supports HMDs compatible with Oculus and OpenHMD. Output is stereoscopic or monoscopic visual.	Data recording and export require custom implementation in Python.	No advanced features reported.
PTVR	Environment and logic are defined via Python scripting, using an event-callback architecture.	Supports HMDs, VR controllers, and keyboards. Integrates with eye-tracking via SRanipal SDK.	Records event-driven logs with timestamps and optional head/gaze data. High timing precision (~2ms of desired duration).	Can import and animate complex 3D characters.

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Table 2 (continued)

Toolbox Name	Experiment Design	Hardware & I/O	Data Handling	Advanced Features
RemoteLab	A Unity toolkit. Logic is defined via C# scripting and custom editor UIs.	Supports HMDs and webcams. Output is a shared VR scene and video conference audio. Integrates with OBS.	Records virtual scene state, questionnaires, and video/audio streams. Exports to CSV and Protocol Buffers.	Has a researcher avatar and is built for multi-user networking.
ROVER	An overlay tool for questionnaires. Logic is configured via a no-code interface.	Supports HMDs and Bluetooth LE heart rate sensors. Output is a visual overlay with audio/haptic feedback. Integrates with SteamVR.	Records questionnaire answers, interactions, and heart rate. Exports to CSV.	Has an instructor avatar and supports partial instructor-participant interaction.
SALLO	A Unity package. Logic is defined via C# scripting for use with tools like UXF.	Supports HMDs, controllers, and K/M. Output is visual and spatialized audio. Integrates with UXF.	Data logging is dependent on the external tool used (e.g., UXF).	No advanced features reported.
SceneExplorer	A visualization tool for existing Unity/Unreal scenes. No logic definition.	Supports HMDs with eye-tracking and biometric sensors.	Records eye-tracking, biometrics, and user actions.	Replays user avatars and supports cloud-based sharing.
SkyBXF	Modding of “The Elder Scrolls V: Skyrim VR” via the Creation Kit. Logic defined via GUI configuration.	Supports HMDs, controllers, and K/M.	Records participant info and questionnaire data. Exports to JSON and CSV.	Uses avatars from the underlying game.
SNaP Framework	A Virtools GUI. Logic is defined via XML configuration files and Python scripting.	Supports trackers, wands, EEG, and fMRI devices. Output is visual and auditory.	Records participant path, camera bitmaps, and behavioral results.	Supports a participant avatar.
Synapse-Unity API	Environment and logic defined via C# scripting and a GUI.	Supports intracranial electrodes for input and provides visual, audio, and direct electrical stimulation output. Integrates with the TDT neurophysiology platform.	Records neural signals, stimulation parameters, and event data. Exports to JSON and UDP formats with high timing precision (6.8ms avg latency).	No avatar, multi-user, or cross-species support reported.
Toggle Toolkit	Environment created in the Unity Editor. Logic is defined via the GUI Inspector or C# scripting.	Supports K/M, controllers, and HMDs. Integrates with external programs and eye-tracking hardware.	Records user movement, interface use, and trigger activations with timestamps. Exports to TXT.	Supports a user avatar.
UXF	Uses existing Unity scenes. Logic defined via C# scripting and JSON configuration files.	Supports any device compatible with Unity.	Records behavioral and continuous data (position/rotation). Exports to CSV. Timing is based on display refresh rate.	Supports a participant avatar and has cloud-based features for remote experiments.
vexptoolbox	Uses the Vizard GUI editor and Python scripting. Logic defined via Python and CSV configuration files.	Supports any Vizard-compatible hardware.	Records discrete trial results and continuous kinematics/eye movements. Exports to JSON and CSV.	Supports avatars via Vizard.
Vizard	Environment and logic are defined via Python scripting and a 3D scene editor.	Supports extensive hardware including HMDs, motion/eye tracking, and haptics.	Data recording and export are user-defined via Python scripting. Reports moderate visual/auditory lag.	Supports avatars, multi-user applications, and has partial cross-species support.
VNT	A tile-based level generator in Unity. Logic is defined via a C# state machine.	Supports K/M, adaptable for eye-trackers. Integrates with UXF for data logging.	Data logging is dependent on the external tool used (e.g., UXF).	Supports avatars and is designed for distributed remote experiments.
VRAT	A pre-built “Virtual Classroom.” Logic is configurable by the instructor.	Supports HTC VIVE HMDs. Integrates with VisiTor cognitive architecture.	Records cognitive and biometric data, attention, and motion. Exports a comprehensive evaluation report.	Supports student avatars and multi-user training.
VREX	A GUI for procedural room generation. Logic is defined via GUI menus and templates.	Supports HMDs, gamepads, and keyboards. Integrates with other peripherals via the Unity interface.	Records behavioral data, scores, and events with timestamps. States millisecond precision.	No primary focus on avatars.
VRSTK	Uses sample scenes in Unity. Logic is defined via a Unity framework (implied C#).	Supports motion capture, eye-tracking, and physiological sensors. Integrates with EEG and Arduino.	Records all sensor data, audio/video, and interactions. Exports to JSON and CSV.	Supports participant/operator avatars and is designed for social VE.

API: Application Programming Interface. **ASIO:** Audio Stream Input/Output. **CSV:** Comma-Separated Values. **EEG:** Electroencephalography. **FSM:** Finite State Machine. **fMRI:** Functional Magnetic Resonance Imaging. **GUI:** Graphical User Interface. **HMD:** Head-Mounted Display. **HR:** Heart Rate. **I/O:** Input/Output. **JSON:** JavaScript Object Notation. **JS:** JavaScript. **K/M:** Keyboard/Mouse. **LSL:** Lab Streaming Layer. **N/A:** Not Applicable. **NPC:** Non-Player Character. **OBS:** Open Broadcaster Software. **TCP:** Transmission Control Protocol. **TDT:** Tucker-Davis Technologies (a neurophysiology hardware company). **TXT:** Text File. **UDP:** User Datagram Protocol. **UDK:** Unreal Development Kit. **UI:** User Interface. **VE:** Virtual Environment. **VR:** Virtual Reality. **XDF:** Extensible Data Format.

editor. Others are hybrid systems that combine GUI or configuration files for structure with scripting for more advanced customization. For instance, toolboxes like NavWell and NeuroVirtual 3D are completely GUI-based toolboxes that anyone can use after a gentle learning curve. While bmlTUX and OpenMaze use a mixture of GUI or configuration files for structure and scripting for advanced customization.

3.3.2. Hardware and I/O support

We tried to include every toolbox that are compatible with major VR HMDs. Beyond standard controllers, a significant number integrate with specialized hardware; 38% (13/34) explicitly support eye-trackers, including Attention Atlas (AA) and PLUME, and 32% (11/34) support physiological sensors or EEG systems, a key feature in toolboxes such as BARGAIN and VRSTK. Output modalities were primarily visual and auditory, with several platforms like Attention Atlas and Ouvrai also supporting haptic feedback. And some platforms like SALLO and Ouvrai are leveraging the spatial audio capabilities of their underlying game

engines.

3.3.3. Data handling

The toolboxes generally provide robust data recording capabilities. The most commonly recorded data types were positional/kinematic data and behavioral data (e.g., response times, user inputs), which were central to nearly all platforms. Toolboxes like UXF and Ouvrai are designed to log the position and rotation of any object in the scene at the display's refresh rate. CSV was the most common data export format was, offered by at least 38% of toolboxes (13/34), valued for its easy to use and integration with a lot of analysis software such as R and Python. JSON was another common format, used by 21% (7/34) for exporting trial configurations and results, as used in bmlTUX. Other toolboxes utilize custom, highly performant formats to handle data; for instance, PLUME uses a modular.PLM format that can be converted to CSV. A critical feature for many research paradigms is temporal precision; 26% of the reviewed toolboxes (9/34) made specific claims of high-precision

timing (e.g., millisecond accuracy or better), including PLUME (100ns clock accuracy) and PTVR (~2ms trial accuracy).

3.3.4. Advanced features

A significant number of toolboxes offered features extending beyond single-user experiments. Support for avatars or agents was very common, found in 71% of the toolboxes (24/34). This ranged from simple participant representations to controllable non-player Characters (NPCs) in BARGAIN or imported animated characters in PTVR. Almost half of the toolboxes (15/34, 44%) provided some kind of multi-user or cloud features, enabling remote collaboration and data collection, as seen in RemoteLab and UXF. Integration with external hardware and software was a key feature for many, with PLUME and OMVR leveraging Lab Streaming Layer (LSL) for synchronizing with a wide array of sensors. Finally, while most toolboxes were designed for human participants, a small subset (3/34, 9%) explicitly noted support for cross-species research, including BonVision and the Doucet_2016 Toolbox.

3.4. Usability and researcher experience

Table 3 provides a summary of the usability and extensibility of the reviewed toolboxes, focusing on the human factors that influence the researcher's experience.

3.4.1. Ease of use and learning curve

The ease of use is largely determined by the primary interaction method. A significant portion of the toolboxes (16/34, 47%) are primarily GUI-based, designed to lower the technical barrier for researchers with minimal programming experience. These tools, such as bmlTUX and OpenMaze, are consistently reported as having a “Gentle” learning curve. Conversely, 35% of the toolboxes (12/34) are scripting-required, demanding proficiency in languages like C# or Python. These platforms, including psychXR and Vizard, offer maximum control but are associated with a “Steep” learning curve. The remaining toolboxes offer a hybrid approach, combining a GUI for basic setup with scripting for advanced customization.

3.4.2. Flexibility and extensibility

The flexibility of a toolbox is strongly linked to its licensing and design philosophy. The vast majority of toolboxes with “High” extensibility (24/28, 86%) are open-source, allowing researchers to freely modify and extend the source code to fit their specific experimental needs. Proprietary solutions like CyberSession can also be highly extensible through their scripting APIs, but do not allow for modification of the core engine. Toolboxes with “Limited” or “Low” extensibility, such as NeuroVR 2 and luVRe, are typically designed as fixed platforms or stimulus sets that are not intended to be fundamentally altered by the end-user.

3.5. Application in psychological and behavioral research

Table 4 details the broad range of applications for the reviewed VR toolboxes, demonstrating their impact across numerous domains of psychological and behavioral science.

3.5.1. Research domains

The most frequently studied domain is spatial navigation and memory, which is the primary focus of at least 10 of the 34 toolboxes, including Landmarks, OpenMaze, and SNaP Framework. Other prominent domains include vision and perception (e.g., bmlTUX, PTVR), motor control and learning (e.g., Ouvrai, MovementVR), and various areas of clinical psychology, such as phobia treatment (CyberSession) and post-stroke rehabilitation (NeuroVirtual 3D).

3.5.2. Experimental paradigms

The toolboxes have been used to implement a wide variety of classic

Table 3

VR toolbox usability and extensibility.

Toolbox Name	Primary Interaction Method ^a	Learning Curve ^b	Extensibility ^c
Attention Atlas (AA)	Hybrid	Gentle-to-Moderate	High (Open-source)
BARGAIN	Hybrid	Moderate	High (Open-source)
bmlTUX	GUI-Based	Gentle	High (Open-source)
BonVision	GUI-Based	Gentle-to-Moderate	High (Open-source)
CyberSession	Scripting-Required	Moderate-to-Steep	High (Proprietary)
Doucet_2016 Toolbox	Scripting-Required	Gentle	High (Custom code)
Landmarks	Hybrid	Gentle-to-Moderate	High (Open-source)
luVRe	N/A (Stimulus Set)	Gentle	Low (Fixed content)
MovementVR	GUI-Based	Gentle-to-Moderate	High (Open-source)
NavWell	GUI-Based	Gentle	Moderate (Customizable)
NeuroVirtual 3D	GUI-Based	Steep (for complex tasks)	Limited (Fixed features)
NeuroVR 2	GUI-Based	Gentle	Limited (Fixed features)
Okada et al. Paradigm	N/A (Pre-built Paradigm)	Gentle	Low (Fixed paradigm)
OMVR	GUI-Based	Gentle	Moderate (Proprietary)
OpenMaze	GUI-Based	Gentle	High (Open-source)
Ouvrai	Scripting-Required	Moderate	High (Open-source)
PLUME	Hybrid	Gentle	High (Open-source)
psychXR	Scripting-Required	Steep	High (Open-source)
PTVR	Hybrid	Gentle	High (Open-source)
RemoteLab	Hybrid	Gentle-to-Moderate	High (Open-source)
ROVER	GUI-Based	Gentle	High (Open-source)
SALLO	Hybrid	Gentle	High (Open-source)
SceneExplorer	GUI-Based	Gentle	Limited (Proprietary)
SkyBFX	GUI-Based	Gentle	High (Game Mod)
SNaP Framework	Hybrid	Gentle	High (Custom code)
Synapse-Unity API	Scripting-Required	Steep	High (Open-source)
Toggle Toolkit	GUI-Based	Gentle	High (Open-source)
UXF	Scripting-Required	Moderate-to-Steep	High (Open-source)
vexptoolbox	Scripting-Required	Moderate	High (Open-source)
Vizard (Platform)	Scripting-Required	Steep	Very High (Proprietary)
VNT	Scripting-Required	Moderate-to-Steep	High (Open-source)
VRAT	GUI-Based	Gentle	Low (Fixed features)
VREX	GUI-Based	Gentle	Limited (Fixed features)
VRSTK	Hybrid	Gentle-to-Moderate	High (Open-source)

^a **Primary Interaction Method:** **GUI-Based:** Primarily operated through a graphical user interface; minimal or no coding required for core functionality. **Scripting-Required:** Core functionality requires programming knowledge (e.g., C#, Python). **Hybrid:** Combines a GUI for basic setup with scripting required for advanced customization.

^b **Learning Curve: Gentle:** Can be used effectively with minimal instruction. **Moderate:** Requires reading documentation and working through tutorials. **Steep:** Requires significant programming knowledge and/or familiarity with the underlying engine. **Note:** usability scores are derived from synthesized evidence in the literature and official documentation rather than direct benchmarking.

^c **Extensibility: High:** Open-source and/or designed with a scripting API for extensive modification. **Moderate:** Allows for some customization of parameters but not core functionality. **Limited/Low:** Designed with fixed features and is not intended to be modified.

and novel experimental paradigms. Several toolboxes, such as Doucet_2016 and NavWell, provide implementations of the Morris water maze. Paradigms for studying attention and memory are also common, with VREX offering built-in templates for change blindness and false memory tasks, and luVRe replicating the “Invisible Gorilla” experiment. In clinical domains, CyberSession is used for exposure therapy scenarios like public speaking simulations, while BARGAIN employs a moral decision-making game to study socio-moral development.

3.5.3. Populations and methodological integration

While healthy adults and university students are the most common target populations, many toolboxes are designed for or have been applied to diverse groups. A significant number of toolboxes are used with clinical populations, including stroke patients (Attention Atlas), individuals with anxiety disorders (CyberSession), and those with Alzheimer's disease (SNaP Framework). Furthermore, a subset of toolboxes, including BonVision and Doucet_2016, is explicitly designed for cross-species research with animal models. Methodologically, integration with other scientific instruments is a key feature. EEG is the most commonly integrated tool, mentioned for 11 toolboxes, followed by eye-tracking, which is integrated into 10 of the reviewed toolboxes. Several also support fMRI (CyberSession, OpenMaze, SNaP Framework) and various other psychophysiological sensors.

3.6. Reported advantages and limitations

Table 5 provides a summary of the reported advantages and limitations for the reviewed VR toolboxes, categorized by their underlying technology or primary function.

3.6.1. Common advantages

A primary advantage highlighted across the literature is the ability to achieve a high degree of ecological validity and presence, allowing researchers to simulate real-world or high-risk conditions in a controlled, safe, and reproducible manner. For example, SceneExplorer is recognized for providing realistic environments for risk-laden scenarios. For researchers with limited programming experience, a key benefit offered by many toolboxes is the reduction of technical barriers. Tools like NeuroVR 2 and OpenMaze are praised for their GUI-based or no-code approaches, enabling the creation of complex experiments without extensive scripting.

Flexibility and modularity are also common strengths; toolboxes such as PLUME and VNT provide modular components that can be adapted and reused, accelerating hypothesis testing. This is often coupled with the benefits of open-source collaboration, where platforms like PLUME foster community development and resource sharing. For scripting-based frameworks like PTVR and Ouvrai, advantages include rapid, reproducible experiment creation and robust support for remote, high-frequency data collection.

3.6.2. Common limitations

The most predominant challenge reported across the literature is the significant technical overhead. Many toolboxes, especially those built on game engines, require a steep learning curve and proficiency in scripting languages like C#. As noted for Landmarks and UXF, users may lack the technical skills to modify templates for highly specific needs, and

platforms like Remotelab can present substantial technical hurdles during setup.

Another common limitation is timing precision. While some toolboxes like NeuroVirtual 3D report millisecond precision, others, such as bmlTUX and OpenMaze, are structurally limited by the engine's main-thread update rate, which is tied to the application's framerate. While it is technically possible to circumvent this by utilizing parallel threading to decouple data logging from rendering performance, such advanced architectures are rarely implemented in “out-of-the-box” toolboxes. Critically, the literature notes that empirical evidence to validate timing claims is often unavailable or anecdotal, complicating the selection process for high-precision cognitive tasks.

Accessibility and sustainability also remain inconsistent. Features for individuals with sensory or motor impairments are rarely standardized. Finally, reliance on proprietary software and evolving licensing models presents a recurring barrier, with commercial licenses for platforms like Vizard and SceneExplorer or dependencies on specific games, as with SkyBXF, creating prohibitive cost or access barriers for many institutions.

4. Discussion

This scoping review has systematically charted the landscape of 34 unique software tools designed for executing behavioral, cognitive, and neuroscientific research within virtual reality. The results show that the ecosystem is vibrant, innovative, and fragmented, and it has grown quickly, with over 70% of the identified toolboxes found since 2020. This growth is mostly happening in North America and Europe, which are home to many major academic research hubs. Our analysis shows a number of important trade-offs that researchers need to make and points to key trends that will affect the future of immersive science.

4.1. Foundational platforms: the power and complexity of commercial game engines

4.1.1. Unity as the unofficial standard in academic research

A systematic review of VR research software must begin with the commercial game engines that constitute the bedrock of the ecosystem. The Unity engine has become the de facto standard for academic research; however, its accessibility is less a result of an inherently flat learning curve and more a reflection of its unrivaled support ecosystem. For researchers with foundational experience in 3D modeling or programming, Unity offers a versatile framework supported by an abundance of community-driven tutorials and troubleshooting forums that lower the barrier to entry compared to other game engines. Unity gives researchers access to a set of advanced tools that were first made for professional game development. These include a powerful rendering engine for making environments look realistic, a strong physics system for simulating how objects interact, and a lot of animation tools.

A key reason in its widespread adoption is its unmatched ability to work cross-platform. You can use a single Unity project and deploy it across a vast range of systems, like desktop computers, mobile devices, and, most importantly, all major VR headsets. This flexibility lets researchers to design experiments that can be run in different modalities. Unity's huge Asset Store offers a marketplace of pre-made 3D models and code snippets, which can dramatically accelerate the prototyping of virtual environments, a significant benefit for research labs with limited resources for content creation. C# is the engine's main scripting language. It is a modern, object-oriented language that is thought to be fairly accessible for those with some programming background.

4.1.2. The “Researcher's tax”: limitations of game engines for scientific inquiry

Even though it's powerful, using a general-purpose game engine for specialized scientific applications has a big cost that is often not fully understood: the “researcher's tax” that comes with development time

Table 4
VR toolbox applications in research.

Toolbox Name	Research Domains	Key Paradigms	Target Populations	Methodological Integration
Attention Atlas (AA)	Visuospatial attention Unilateral spatial neglect Cognitive impairment	3D visual search (target/distractor)	Stroke patients Brain injury patients Healthy controls	Eye-tracking: To map visuospatial attention.
BARGAIN	Affective state Emotion Socio-moral development	Moral decision-making game ("Dilemmas_VR")	Healthy adults	EEG: To assess emotional state.
bmlTUX	Vision & perception 3D perception Stereopsis	Multifactorial experimental designs Adaptive staircase methods	University students	Note: Authors suggest other tools for high-precision timing.
BonVision	Visual neuroscience Animal behavior Spatial navigation	Receptive field mapping Virtual platform navigation Looming dot stimulus	Humans Mice	Electrophysiology Deep neural networks (for pose estimation)
CyberSession	Psychology Neuroscience Clinical Psychology	Public speaking simulation Driving simulation Phobia treatment	Healthy adults Clinical groups	EEG fMRI Eye-tracking Psychophysiology
Doucet_2016	Spatial navigation Memory Contextual learning	Morris water maze (analogue) Radial arm maze Contextual learning task	Rodents Non-human primates Humans	Eye-tracking Electrophysiology
Landmarks	Spatial Navigation Learning & Memory	Navigation tasks Route-learning Scene/orientation pointing	Healthy adults	Eye-tracking (Tobii Pro) Omnidirectional treadmills
luVRe	Emotion & Motivation Autobiographical memory Attention	Emotional response to videos Free/cued recall tasks Change blindness ("Invisible Gorilla")	Healthy adults	EEG: To measure frontal alpha asymmetries.
MovementVR	Motor control & learning Motor rehabilitation	Naturalistic reaching/lifting tasks Motor adaptation (gain perturbation) Reinforcement learning	General human subjects Potential clinical groups (e.g., stroke, cerebral palsy)	Integrated hand-tracking Kinematic data analysis
NavWell	Spatial navigation Spatial memory Spatial learning	Morris water maze paradigms (reference, working, cued)	Healthy young & older adults Potential clinical groups	Behavioral measures Standard cognitive tasks
NeuroVirtual 3D	Neuropsychology Psychometrics Spatial memory Post-stroke rehab	Spatial ability assessment Unilateral Spatial Neglect tasks	Patients (Stroke, Parkinson's, OCD) Elderly participants	Biosensors (Zephyr, StarStim) Microsoft Kinect
NeuroVR 2	Clinical psychology Behavioral health Cognitive rehab	Exposure therapy scenarios Virtual Multiple Errands Test Addiction cue assessment	Clinical groups (e.g., anxiety, obesity)	Future goal: Eye-tracking, Psychophysiology.
Okada et al. Paradigm	Episodic memory Scene recognition Déjà vu & Familiarity Spatial cognition	Virtual tour (study/test phases) Scene recall & familiarity detection Turn prediction task	Healthy adults Potential clinical use	fMRI Electrophysiology Intracranial recordings
OMVR	Hearing science Spatial hearing Multimodal integration	Spatial Speech in Noise (SSiN) tests Sound localization tasks	Normal-hearing adults Cochlear implant users	Pupillometry (PupilLabs) EEG (g.Tec) via LSL
OpenMaze	Spatial navigation Spatial memory & learning	Object-place learning Wayfinding & foraging tasks	University students Potential clinical use (e.g., preclinical Alzheimer's)	EEG fMRI (noted as a strong option)
Ouvrai	Sensorimotor functions Motor control & learning Perception	Visuomotor rotation Hand/finger tracking Physics-based interaction	Healthy adults (via crowdsourcing)	High-resolution kinematics
PLUME	User behavior Quality of Experience (QoE)	"Easter egg" searching task in an interactive environment	General human subjects	Physiology (Heart rate via LSL)
psychXR PTVR	Learning analytics General Psychophysics Visual perception Visual search	User-defined via Python scripting Standardized reading test (MNREAD) 3D visual search	Healthy adults Low-vision patients	Python-compatible libraries Eye-tracking (Vive Sranipal)
RemoteLab	Low-vision reading Human-computer interaction Remote user studies	Head-contingent pointing In-VR questionnaires (NASA-TLX, SUS) Researcher-guided remote tasks	Remote, diverse populations	Video conferencing (Zoom, Teams) OBS
ROVER	Health psychology User experience Cognition/emotion	Rhythm-based exergaming (e.g., Beat Saber) with in-VR questionnaires	Older adults VR beginners	Heart rate sensors (Bluetooth LE)
SALLO	Spatial orientation Multimodal perception Spatial reasoning	Left-right discrimination Space bisection tasks	Healthy sighted individuals	Inertial measurement units (IMUs)
SceneExplorer	Post-hoc analysis (any domain)	N/A (analysis tool)	Any population	Eye-tracking Biometrics
SkyBXF	Cybersickness (VIMS)	Virtual walk task with Simulator Sickness Questionnaire (SSQ)	Healthy adults	Psychophysiology (via SSQ)

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Table 4 (continued)

Toolbox Name	Research Domains	Key Paradigms	Target Populations	Methodological Integration
SNaP Framework	Spatial navigation	Battery of spatial navigation paradigms	Elderly Alzheimer's/dementia patients	EEG fMRI
Synapse-Unity API	Brain-computer interface (BCI) Seizure localization Neural stimulation	Closed-loop BCI task (neural signals control virtual object)	Patients with intracranial electrodes	Intracranial electrocorticography (ECoG) Stereo-electroencephalography (sEEG)
Toggle Toolkit	Behavioral analysis Spatial cognition Attention	Evacuation behavior analysis Foreground-background tasks	Healthy adults	Eye-tracking (gaze-based triggers)
UXF	Interceptive timing Postural control	Virtual "swinging room" paradigm	Adults & children	Head-tracking (to measure postural sway)
vexptoolbox	Spatial processing Cognitive/perceptual processes	Pro- and anti-reach tasks Goal-directed reaching	Healthy adults	Eye-tracking EEG
VNT	Spatial navigation Spatial cognition & memory	Triangle completion (homing) Trap-lining (route optimization)	Healthy adults	Eye-tracking UXF framework
Vizard	Spatial perception Presence & embodiment Stimulus timing validation	Pro- and anti-reach tasks Stimulus presentation tests	Healthy adults	Motion tracking (OptiTrack, Vicon) Eye-tracking EEG
VRAT	Attention Education	Virtual Classroom tasks Attention tests	Students	Cognitive & biometric data
VREX	Perception & Attention Cognition & Memory	Change blindness (template) False memory (template)	General psychology participants	Extensible via Unity, but not a primary focus.
VRSTK	Human factors Social interaction Communication	Social Fear Conditioning paradigm	General human participants	EEG Psychophysiology Motion capture Eye-tracking

and possible threats to scientific validity. The main problem is that game engines like Unity aren't made for research. They don't have the basic architectural parts needed for controlled behavioral experiments, like built-in structures for managing sessions, blocks, and trials; advanced data logging systems; and accurate tools for stimulus control. As a result, researchers have to create these important parts from scratch, which is not very efficient and results in codebases that are unique and hard to replicate.

The most important limitation may be the difficulty of getting precise temporal control within general-purpose engines. In domains such as neuroscience and psychophysics, where millisecond-level precision is non-negotiable, Unity's default execution logic is intrinsically coupled to the rendering loop (the display's frame rate), rendering stimulus onset and data logging susceptible to framerate-induced variability. While parallel threading offers a technically viable pathway to decouple time-critical logic from visual performance, its implementation demands an architectural sophistication that exceeds the baseline proficiency of most research teams. This necessitates a difficult trade-off: utilizing specialized platforms like Vizard, which offer high-precision timing out-of-the-box but lack Unity's extensive community support, or investing heavily in custom engineering. Ultimately, the substantial technical overhead, comprising both the steep learning curve of the C# API and the complexities of the Unity environment; imposes an innovation tax. This burden forces research groups without dedicated programming staff to prioritize technical troubleshooting over scientific inquiry, effectively diverting resources away from the core pursuit of discovery.

4.1.3. The open-source dimension

The relationship between Unity and the academic research community is further complicated by its licensing model. Unity itself is proprietary software, but its "freemium" model makes it highly accessible to non-commercial academic labs. This proprietary foundation has given rise to a vibrant ecosystem of open-source research toolboxes built on top of Unity. These toolboxes are designed to fill the gaps left by the core engine—providing the experimental structures and hardware integrations that researchers need. This creates a hybrid model where a proprietary platform serves as the host for a community-driven, open-source layer of scientific tools.

The dominance of Unity is a path-dependent phenomenon driven by a powerful feedback loop. Its initial accessibility led to its adoption by early academic researchers. These pioneers, in turn, built and shared the first generation of open-source toolboxes. The existence of this "scaffolding" then made Unity an even more attractive choice for the next wave of researchers. This network effect has cemented Unity's position as the central platform in the field, even as its inherent scientific limitations remain a significant concern.

It should be noted, however, that Unity's dominance is currently facing its first significant challenge. While its 'Personal' tier remains a gateway for individual researchers, the **\$200,000 funding threshold** for forced upgrades to Pro seats has made the engine less affordable for mid-sized academic labs. This economic barrier, combined with the desire for more transparent and 'locked-in' experimental codebases, is fueling a growing interest in open-source engines like Godot, which provides a similar C#-friendly environment without the institutional overhead.

4.2. Engine-based research toolboxes and frameworks

The limitations of using a raw game engine have given rise to a diverse ecosystem of toolboxes, frameworks, and plugins. These tools sit on top of foundational platforms like Unity, providing the necessary functionalities for experimental research.

The identified tools can be classified into a functional taxonomy: (1) **General Experiment Frameworks** provide the fundamental "backbone" for a behavioral experiment (e.g., sessions, blocks, trials) and robust systems for data logging. (2) **Domain-Specific Toolkits** are specialized packages designed for a particular paradigm, such as spatial navigation or motor learning. (3) **Hardware/Software Integration APIs** connect the virtual environment with external systems like EEG, fMRI, or eye-trackers.

4.2.1. General Experiment Frameworks: building the experimental backbone

General frameworks transform a game engine into a viable scientific instrument. The **Unity Experiment Framework (UXF)** is a powerful plugin that provides a comprehensive structure for behavioral experiments. Its strength lies in its extensive set of C# classes, giving

Table 5
Summary of reported advantages and limitations.

Toolbox Name (or Category)	Reported Advantages	Reported Limitations/ Challenges
<i>General Game Engines</i>		
Unity, Unreal	Graphics & Physics: Powerful rendering, physics, and animation. Cross-Platform: Extensive support for multiple devices. Community & Ecosystem: Large communities and plugin support. Flexibility: Scripting support (C#, C++).	Lack of Research Tools: No built-in features for data output or precise timing. High Technical Barrier: Requires custom coding and solid knowledge. Reproducibility Issues: Codebases can be idiosyncratic, hindering replication.
<i>Toolboxes Built on Game Engines</i>		
BARGAIN (Unity)	Affective Adaptation: Adapts game elements based on player EEG data. Integration: Can be embedded as a package into any Unity game.	Technical Barrier: Requires Unity setup and device integration. Documentation: Described as evolving. Customization: Advanced changes require C# scripting.
bmlTUX (Unity)	GUI for Design: Simplifies combinatorics and randomization. Accessibility: Enables experiment creation with minimal coding. Customizable: Suitable for both novices and experts.	Timing Precision: Limited to Unity's update rate. Design Limitations: Less suited for adaptive or non-factorial designs. Hardware Integration: Limited built-in support for external hardware.
Doucet 2016 (Unreal)	Cross-Species: Supports both rodents and humans. Timing Precision: Aims for millisecond timing for electrophysiology. Integration: Works with Matlab/Python for experiment control.	Outdated Technology: Based on the older Unreal Engine 3 (UDK). Timing Jitter: Achieving millisecond precision is challenging ($\sim \pm 3$ ms jitter).
Landmarks (Unity)	Integrated GUI: Simplifies experiment setup. EEG Integration: Out-of-the-box tools for VR/EEG studies. Centralized Framework: Manages process from design to data. Active Maintenance: Supported by a wiki and active repository.	Rigidity: Autogenerated tasks can be difficult to modify. Learning Curve: Requires C# for advanced customization. Dependency: Requires Unity installation.
MovementVR (Unity)	Accessibility & No Coding: Open-source, intuitive GUI to create realistic motor learning tasks without programming. Ecological Validity: physics-based hand-object interaction and real-time hand tracking. Motor Learning Focus: Natively supports motor adaptation (gain perturbation) and reinforcement learning, making it ideal for research and rehabilitation. Affordability: Runs on low-cost, consumer-grade VR headsets.	Task Specificity: Primarily designed for a specific "lifting task"; creating other task types requires modifying the backend Unity code. Limited Perturbations: Built-in motor adaptation is limited to gain perturbations. Tracking Constraints: Motion tracking is limited to the wrist and forearm and does not support upper arm tracking. Environmental Dependency: Hand tracking performance relies on optimal lighting conditions
OpenMaze (Unity)	Asset Creation: Users must create or import their own 3D assets. Hardware Support: Advanced VR hardware requires Unity add-ons. Timing Precision: Tied to Unity's update rate.	Asset Creation: Users must create or import their own 3D assets. Hardware Support: Advanced VR hardware requires Unity add-ons. Timing Precision: Tied to Unity's update rate.
PLUME (Unity)	Modularity: Plugins for data capture and annotation. Standardization: Promotes	Customization: May require coding for highly specialized paradigms.

Table 5 (continued)

Toolbox Name (or Category)	Reported Advantages	Reported Limitations/ Challenges
	standardized data formats. Reproducibility: Designed for data sharing and replication. Ease of Use: The recorder is plug-and-play and can be added to projects with minimal configuration. Advanced Visualization: A standalone viewer provides interactive 3D replay and in-context visualizations	Single-User Analysis: The analysis modules are currently limited to single-user records. Viewer Performance: The standalone viewer application may have performance issues when handling very large recordings.
RemoteLab (Unity)	Multi-User Support: Designed for remote, synchronized experiments. Researcher Control: Allows interaction as an avatar and multiple viewpoints. Data Collection: Supports real-time data gathering.	Technical Overhead: Requires Unity setup and web deployment. Network Latency: Potential for lag in remote sessions. Support: Described as evolving for complex paradigms.
SALLO (Unity)	Templates: Provides modular templates for stimuli, tasks, and methods. Accessibility: Lowers technical barriers for spatial cognition research. Integration: Designed to work with UXF.	Niche Focus: Limited to spatial orientation paradigms. Dependency: Requires Python 2.7 for certain algorithms. Technical Barrier: Requires Unity setup and some scripting.
SkyBXF (SkyrimVR)	Rapid Setup: Designed for users with minimal programming/3D design skills. Asset Reusability: Leverages existing assets from SkyrimVR. Familiar Environment: Uses a well-known commercial game world.	Engine Dependency: Relies on a proprietary game and its engine. Limited Scope: Restricted to environments/mechanics available in Skyrim.
Synapse-Unity API (Unity)	BCI Integration: Enables bidirectional communication between VR and neural data. Real-Time Feedback: Supports closed-loop neural feedback. Rapid Development: Speeds up task creation in Unity.	Technical Barrier: Advanced customization requires modding knowledge. Niche Focus: Primarily for labs using TDT neurophysiology systems. Data Bottlenecks: UDP interface has bandwidth constraints.
Toggle Toolkit (Unity)	Dynamic Events: Allows object changes based on triggers (e.g., gaze, collision). Accessibility: High customizability with minimal coding. Event Logging: Built-in logging for statistical analysis.	Hardware Dependency: Requires specific TDT hardware. Platform Lock-in: Limited to Unity-based environments. Paradigm Support: Not all experimental designs are supported out-of-the-box.
UXF (Unity)	Experiment Structure: Provides a comprehensive session/block/trial lifecycle. Data Collection: Robust tools for tracking objects and saving data. Reproducibility: Enables rapid and reproducible development.	Technical Barrier: Requires Unity and C# scripting knowledge. No Stimulus Tools: Relies on user-created Unity assets.
VNT (Unity)	Modularity: Interchangeable tools for spatial navigation. Code Reusability: Components can be adapted and shared. Integration: Can be used with existing tools like UXF.	Technical Barrier: Requires Unity setup and C# scripting.
VREX (Unity)	Ease of Use: GUI simplifies stimulus and environment	Limited Scope: Interface is restricted to "apartment-

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Table 5 (continued)

Toolbox Name (or Category)	Reported Advantages	Reported Limitations/Challenges
VRSTK (Unity)	creation. Templates: Provides scripts for common paradigms (e.g., change blindness). Timing Precision: Reports millisecond-level precision. Holistic Data: Real-time tracking of movement, gaze, EEG, etc. Operator Control: Allows live control and scene replay. Reproducibility: Standardized data export and pre-built modules. Multiplayer Support: Designed for social VR experiments.	like" settings. Hardware Dependency: Designed for older Oculus Rift headsets. Lack of Flexibility: Key bindings are hardcoded. Signal Interference: VR equipment can be a challenge for physiological signals. Feature Completeness: Some advanced features are still in development. Documentation: Described as evolving.
Okada et al. Paradigm (Unity)	Stimulus Availability: Provides a large, ready-to-use set of scenes. Methodological Rigor: Enables controlled list-learning approach. Versatility: Adaptable for various immersion levels and methods. Open Science: Stimuli and data are publicly available on OSF.	Passive Navigation: Participants do not control their own movement Egocentric Perspective: Does not explore allocentric viewpoints Prompting Difference: Response type differed between 2D and VR versions.
Specialized/Integrated Systems		
Attention Atlas	Specific Assessment: Designed for the assessment of visuospatial attention using engaging, gamified tasks. Precise Mapping: Uses raycasting and eye-tracking for efficient mapping of attention.	Niche Focus: Focuses on a specific cognitive function and population. Hardware Dependency: Requires specific VR hardware and technical setup.
BonVision	Closed-Loop Control: Real-time hardware integration. Graphical Programming: Uses the Bonsai visual language. Extensibility: Can be extended with C# scripting.	Language Dependency: Relies on the Bonsai programming language. Performance: High frame rates can be a challenge for video playback.
CyberSession	Controlled Environments: Enables highly controlled and repeatable experiments with automated scenario control. Hardware Support: Supports a wide range of input devices, including HMDs, motion simulators, and physiological sensors.	Proprietary: Not open-source; requires certified hardware and technical setup. Limited Flexibility: Custom development outside of supported modules is limited.
NavWell	Accessibility: A free, readily downloadable virtual water maze system for desktop or immersive VR. Standardization: Offers "Education" and "Research" versions to help standardize navigational research.	Limited Flexibility: Not well-suited for non-navigation paradigms. Learning Curve: Has an initial learning curve for non-experts.
NeuroVR 2	No Programming Required: A free and open-source platform designed for non-experts, featuring a drag-and-drop editor to customize pre-designed scenarios.	Limited Functionality: Not easily extendable and less flexible than scripting-based tools. Discontinued: This platform has been discontinued.
NeuroVirtual 3D	Device Integration: A free, open-source platform that uses VRPN/TCP-IP to connect with a wide range of devices, such as biosensors for biofeedback. Flexible Design: Uses a drag-	Curve: Has an initial learning curve for non-experts, and complex scene manipulation may be challenging.

Table 5 (continued)

Toolbox Name (or Category)	Reported Advantages	Reported Limitations/Challenges
OMVR	and-drop design for creating highly flexible virtual environments. Hearing Research Focus: A VR-enabled platform specifically for hearing research (e.g., spatial hearing, multimodal sensory integration). Ecological Validity: Enables experiments in realistic 3D soundscapes.	Domain-Specific: Tightly focused on hearing research. Availability: Not publicly available; shared with partners.
SNAP Framework	Hybrid Control: Combines the control of custom systems with the ease of use of pre-packaged systems using XML files. Flexibility: Allows for easy switching between paradigms, input devices (joysticks, EEG, fMRI), and output contexts (monitors, HMDs, CAVEs).	Dependency: Dependent on the proprietary software Virtools. Status: Not open-source or widely distributed. Limited Use: Has limited flexibility for non-navigation paradigms.
VRAT	Real-World Simulation: Simulates real-world activities of daily living (e.g., kitchen tasks) in a realistic 3D environment. Data Collection: Collects kinematic, physiological, and performance data with automated scoring.	Limited Flexibility: Not designed for custom experiment creation. Validation: Current validation is based on small samples.
Scripting-Based Frameworks		
Ouvrai	Remote Studies: Built-in infrastructure for remote hosting and crowdsourcing. Data Quality: Produces high-frequency kinematic data. Open-Source: Avoids strong limitations on customizability.	Technical Barrier: Requires basic scripting and command-line skills. WebXR Limitations: Advanced graphics or device integration may be limited. Cost: Data storage is capped by the free Firebase tier.
psychXR	Low-Level Control: Provides flexible, direct control of device features. Lightweight: Has minimal dependencies. Integration: Works with scientific Python libraries and PsychoPy.	High Technical Barrier: Requires significant programming effort to create experiments. Hardware Support: Can be limited to specific HMDs.
PTVR (Unity backend)	Accessibility: Users do not need Unity knowledge or installation. Rapid Development: Enables reproducible experiments with high-level scripts. Vision Science Tools: Supports gaze/head tracking and perimetric coordinates.	Niche Focus: Primarily on visual perception. Low-Level Control: May offer less control than frameworks like PsychXR. Feature Completeness: Some key features are still in development.
vextoolbox (Vizard)	Rapid Creation: Supports the rapid creation of trial-based designs, randomization, and standardized data export. Data Recording: Features include continuous recording of eye and body movements. Hardware Compatibility: Compatible with any hardware supported by Vizard.	Dependency: Requires the Vizard platform and Python scripting knowledge. Limited GUI: Does not have a graphical interface for experiment design. Timing Accuracy: Timing accuracy is limited by the display's frame rate
Vizard (WorldViz)	Hardware Connectivity: Unparalleled support for motion/eye tracking, haptics. Python-Based: Accessible scripting language.	Commercial: Requires a commercial license. Lack of Experiment Tools: Requires external toolkits for trial-based design.

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Table 5 (continued)

Toolbox Name (or Category)	Reported Advantages	Reported Limitations/Challenges
	Timing Precision: Supports precise millisecond timing.	Hardware Gaps: Does not support EEG or facial expression hardware.
<i>Analysis, Data, & Niche Tools</i>		
luVRe (Data Repository)	Standardized Stimuli: A standardized database of immersive 3D/360° pictures and videos to enhance ecological validity. High Realism: Provides high-realism content for studying emotional and cognitive processes.	Content Only: Does not provide a toolbox or framework for running experiments, only the stimuli. Limited Interaction: Interaction is mostly limited to head-tracking.
ROVER (Questionnaire Tool)	Application Agnostic: Functions as a standalone overlay, allowing questionnaires to be embedded in existing VR applications without code integration. Maintains Immersion: Allows for data collection while maintaining the immersive experience.	Compatibility Issues: Can have compatibility issues with recent VR SDK updates. Limited Scope: Limited to overlay-based questionnaires, not full experiment authoring.
SceneExplorer (Analysis Tool)	Post-Hoc Analysis: 3D replay and analytics of simulation sessions. Data Synchronization: Synchronizes eye tracking, biometrics, and actions. Accessibility: Browser-based access and cloud sharing.	Not an Authoring Tool: Focuses on analysis/replay, not experiment creation. Commercial: Is a commercial tool, which may be a cost barrier. Learning Curve: Requires some technical expertise for setup.

programmers granular control. However, this power comes at the cost of a steep learning curve, requiring solid knowledge of C# scripting (Brookes et al., 2020). In contrast, VREX offers a more accessible, GUI-driven solution. It provides built-in templates for common paradigms like change blindness and simplifies the creation of indoor environments. The trade-off is a significant loss of flexibility (Vasser et al., 2017). bmlTUX occupies a middle ground, with a graphical interface that allows researchers to specify multifactorial experimental designs with minimal coding (Bebko & Troje, 2020). The existence of these distinct frameworks reveals a fundamental tension between abstraction and control. Tools like VREX provide a high level of abstraction to broaden access, but this limits the scope of possible research. Code-heavy frameworks like UXF offer minimal abstraction, providing granular control but demanding significant technical expertise.

4.2.2. Domain-specific toolkits: tailored solutions for common paradigms

The prevalence of toolkits for certain domains highlights the areas where VR has had the most significant impact. spatial navigation and memory are arguably the primary use case for VR in cognitive science, reflected in the sheer number of dedicated toolkits (OpenMaze, Landmarks, NavWell, VNT). This proliferation, however, also points to a significant duplication of effort and a lack of standardization (Alsbury-Nealy et al., 2022; Commins et al., 2020; Müller et al., 2023; Starrett et al., 2021). In visual perception and attention where precise stimulus control is essential, Attention Atlas is a highly specialized project for assessing visuospatial neglect (Norwood et al., 2023), while BonVision is a powerful tool for visual neuroscience that supports cross-species research (Lopes et al., 2021). PTVR represents an innovative hybrid approach, allowing researchers to write experiments in Python while leveraging a Unity engine backend for rendering (Castet et al., 2024).

4.2.3. Hardware/Software Integration: bridging the virtual and physical worlds

Integrating the virtual world with physical data acquisition systems is a critical challenge. Several tools are explicitly designed for this. The Synapse-Unity API provides a dedicated link between Unity and TDT neurophysiology systems for closed-loop brain-computer interface (BCI) experiments (Montag et al., 2021). Frameworks like OpenMaze and VRSTK also report tools for integrating with EEG and fMRI systems (Alsbury-Nealy et al., 2022; Wölfel et al., 2021). The adaptation of Lab Streaming Layer (LSL), an open-source middleware providing standardized synchronization of multiple data streams, by VR toolboxes like OMVR and PLUME represents a significant step towards robust multi-modal integration and greater interoperability (Javerliat et al., 2024; Pedersen et al., 2023).

4.3. The challenge of adoption, impact, and sustainability

A critical question is whether these tools are used beyond the labs that created them. Our analysis suggests that successful adoption is tied to several factors: a permissive open-source license, comprehensive documentation, active maintenance (e.g., on GitHub), and solving a common research problem (Saravanan & Balakrishnan, 2019). Conversely, tools that fail to gain traction are often hindered by proprietary licenses, poor documentation, project abandonment, or being too niche (Eibl & Thurnay, 2023).

It should be noted that this review maps the existence and characteristics of tools, not their scientific or clinical impact. Assessing metrics such as the number of studies that have used a tool, the number of patients tested, or the clinical outcomes achieved would require a separate bibliometric or systematic review. This information is often not available in the primary publication describing the tool, making a direct assessment of impact beyond the scope of this work.

The prevalence of open-source licensing is a major strength, fostering transparency and collaboration. However, it also presents a significant sustainability challenge (Raza & Capretz, 2015). Most academic open-source tools are developed and maintained by a small number of individuals within a single research lab. This creates a precarious situation where the long-term viability of a critical research instrument depends on the continued effort of its original creators, with a high risk of project abandonment (Steinmacher et al., 2015).

4.4. Specialized and integrated systems: turnkey solutions

Distinct from extensible toolboxes built upon general-purpose game engines, another category exists: specialized and integrated systems. These tools function less like development kits and more like complete, turnkey instruments or “appliances.” They often prioritize ease of use, standardization, and reliability over programmatic flexibility. The “appliance” model of research software focuses on providing a complete, out-of-the-box solution that requires minimal technical setup. This approach offers advantages for clinical settings where standardized protocols, reproducibility, and reliability are essential for safe and consistent patient care. The primary trade-off is a significant reduction in flexibility. This highlights a fundamental tension between reproducibility/standardization and flexibility/innovation. This is a well-recognized challenge in the development of VR-based research tools. The accessibility of highly standardized VR platforms enhances the reproducibility of clinical research, but these benefits often come at the cost of reduced flexibility and options for innovation (Parsons, 2015; Rizzo et al., 2018).

A significant portion of specialized systems is targeted at clinical neuropsychology, where VR-based tools have demonstrated robust validity and reliability across a range of cognitive assessments.

Meta-analytic research confirms that appliance-style VR platforms provide both consistent and ecologically valid measures for neuropsychological assessment in clinical populations (Neguț et al., 2016).

NeuroVR 2 exemplifies this model: a free, open-source platform with a user-friendly, drag-and-drop “SceneEditor” allowing non-programmers to customize a library of over 14 pre-designed, real-life scenarios (e.g., a supermarket, an office) for applications in exposure therapy, executive function assessment (e.g., a virtual Multiple Errands Test), and addiction assessment. Experiment logic is defined through a simple “Action & Trigger” system, avoiding any need for scripting. Its primary strength is its accessibility for clinicians and therapists who are not software developers. Its core limitation is the flip side of this same coin: its functionality is limited to the built-in features, and it is not designed to be easily extended via programming, making it unsuitable for researchers who need to create novel or highly customized paradigms (Riva et al., 2011). **NeuroVirtual 3D Evolved** from NeuroVR, maintaining the no-code design while expanding research capabilities for by adding integration with external hardware like biosensors and Microsoft Kinect (Cipresso et al., 2016).

At the other end of the spectrum from free, open-source clinical tools are high-end, proprietary commercial systems. These platforms are typically sold as complete solutions that include software, certified hardware, and professional support, targeting well-funded research labs and industrial applications. **CyberSession** is a long-standing commercial system providing a highly controlled environment for applications like public speaking simulation and phobia treatment. Its key advantages are the professional support, guaranteed hardware compatibility, and robust integration with a wide array of peripherals. The scenarios are controlled via a scripting language, offering a degree of flexibility. However, as a proprietary, closed-source system, it lacks the open extensibility of engine-based toolboxes and requires a significant financial investment (North et al., 1998). **Vizard (WorldViz)** is a major commercial alternative specifically targeted at the academic market. Its key advantages are its Python-based scripting language and its engineering for scientific precision, providing reliable millisecond-level timing (WorldViz, 2023).

4.5. Broader context and future directions

VR-based methods offer significant advantages over traditional laboratory paradigms, most notably a dramatic increase in ecological validity (Parsons, 2015). By immersing participants in realistic, dynamic environments, researchers can study behavior in contexts that better approximate the real world (Negut et al., 2016). This allows for the investigation of complex cognitive processes like spatial navigation and decision-making in ways that are impossible with static, 2D stimuli. The primary disadvantage remains the technical overhead and the potential for methodological confounds like cybersickness (Rizzo et al., 2018).

Several gaps point toward future development priorities. While a few toolboxes support multi-user experiments, this remains a niche capability despite its immense potential for social cognition research (Freeman et al., 2017). Future development in this area will need to address technical challenges like network latency and behavioral synchronization to enable robust studies of social interaction, joint attention, and collaborative problem-solving (Pan & Hamilton, 2018). A striking gap in the current landscape is the near-total absence of AI integration. AI could be a transformative addition, enabling adaptive difficulty that adjusts in real-time to a participant's performance, procedural content generation to create an endless variety of novel environments, and intelligent virtual agents whose behavior is more believable and responsive (Inkarbekov et al., 2023; Zhang et al., 2024).

4.5.1. Beyond VR: the spectrum of XR

While this review focused on immersive VR, the principles and many of the tools discussed are relevant to the broader spectrum of Extended Reality (XR). Future research should explore how these toolboxes can be adapted for Augmented (AR) and Mixed (MR) Reality, which present unique opportunities for studying cognition in real-world contexts (Salatino & Burin, 2024; Thees et al., 2020).

4.6. A decision-making framework for selecting the optimal VR research platform

The optimal choice of software is not absolute but is contingent upon the specific context of the research. This section synthesizes the findings into a practical, decision-making framework. To select the most appropriate platform, researchers should systematically assess their project needs across four key dimensions:

1. What is your research domain and paradigm?
2. What is your team's technical expertise?
3. What are your hardware integration needs?
4. What is your project's sustainability model including the budget and maintenance status?

Table 6 provides a decision-making matrix that maps research scenarios to specific software tools and categories discussed in this review. To assist in software selection, the matrix explicitly outlines the answers to these four key questions for specific categories of tools and software examples.

5. Limitations

While this review provides a comprehensive map of the VR toolbox landscape, it is important to recognize several limitations that frame its conclusions. Perhaps the most significant of these is the study's reliance on secondary data. The evaluation of toolbox features and usability, as summarized in the results, was constructed from peer-reviewed reports and available documentation rather than a direct, hands-on audit of every included platform. As a result, the usability ratings do not represent a standardized, head-to-head technical benchmark; instead, they reflect the reported experiences of users and developers, filtered through the lens of published literature.

The search strategy itself also introduces boundaries. By limiting the scope to English-language literature, the review may have inadvertently excluded robust toolboxes developed and documented in other linguistic contexts, narrowing the global picture. Compounding this is the relentless pace of software development, which means the landscape is inherently fluid. Newer versions of the tools discussed, or significant shifts in their economic models—such as the recent changes to Unity's pricing structure—may have already occurred since the December 2024 search window, subtly altering the practical calculus for developers.

Furthermore, the review synthesizes general trends in usability and performance, but it cannot fully account for the deep entanglement between software and hardware. The user experience of any given toolbox is often contingent on specific configurations, from the particular head-mounted display to the intricacies of GPU requirements, and these nuanced interactions are necessarily flattened in a broad overview. Finally, even with a multi-reviewer approach designed to minimize subjectivity, a thread of interpretive bias is woven into the findings. The thematic analysis of “usability” remains, at its core, an act of interpretation. Looking ahead, future research would benefit greatly from shifting its focus toward primary user-experience studies, employing direct performance metrics to validate and deepen the insights this review has mapped out.

6. Conclusion: synthesizing the landscape and charting future directions

6.1. Summary of key trade-offs

This scoping review has methodically mapped the vast and rapidly changing field of software for VR-based behavioral and cognitive research. The analysis shows that the choosing a software tool is not as easy as selecting the one with the most features. Instead, it is a multi-faceted decision that requires weighing a number of basic and often

Table 6

A decision-making framework for selecting the optimal VR research platform.

Research Domain & Paradigm	Req. Technical Expertise	Hardware Integration	Budget	Maintenance Status	Recommended Tools (Category)
Standardized Clinical Assessment	None (Clinicians/Practitioners)	None or simple biosensors	Free/Open Source	Inactive	NeuroVR 2, NeuroVirtual 3D, NavWell (Specialized Clinical Systems)
Common Behavioral Paradigms	None (Basic Config Files)	None	Free/Open Source	Stable/Inactive	OpenMaze (S), VREX (I) (No-Code Toolkits)
Novel Behavioral Paradigms	Intermediate (Python)	None/Eye-Tracking	Free/Commercial	Active/Inactive	Vizard (A) w/vexptoolbox (I), PTVR (A) (Python-Based Scripting)
Novel Behavioral Paradigms	Advanced (C#/Unity)	Any Unity-compatible	Free/Open Source	Stable/Inactive	UXF (S), VNT (I) (General Experiment Frameworks Code-centric)
Psychophysics/Vision Science	Advanced (Python)	High-precision/Eye-Tracking	Free/Commercial	Active/Stable	Vizard (A), psychXR (S) (High-Precision Scripting)
Multimodal Neuroscience	Advanced (C#/Python)	Complex, multi-channel (LSL)	Free/Open Source	Inactive	VRSTK, OMVR, Synapse-Unity (Integration-Focused Frameworks)
Large-Scale Remote Studies	Intermediate (Web Dev/JS)	consumer VR hardware only	Free/Open Source	Active	Ouvrai (Web-Based VR Platforms)
Rapid Prototyping/Exploratory	Beginner to Intermediate	None	Free/Open Source	Stable/Inactive	Landmarks (I), bmlTUX (S) (GUI-Driven Toolkits)

Abbreviations (A) Active. (S) Stable. (I) Inactive.

conflicting trade-offs. The ecosystem is defined by these critical dichotomies: (1) **Ease of Use** usually comes at the cost of **Flexibility**, tools with graphical user interfaces and no-code solutions lower the barrier to entry, enabling a wider range of researchers to use VR. However, this ease of access usually comes at the cost of experimental flexibility, which limits researchers to pre-defined paradigms. (2) **Proprietary Support vs. Open-Source Collaboration**: Commercial platforms offer the stability of professional support, documentation, and guaranteed maintenance. In contrast, the open-source academic tools foster a collaborative spirit, are free of cost, and offer transparency, but their long-term viability is often precarious. (3) **Graphical Fidelity vs. Scientific Precision**: Modern game engines can make environments that are very realistic and immersive by using the best rendering technology. Yet, these same engines frequently lack the integrated, verifiable, millisecond-level timing precision essential for rigorous research in various fields of neuroscience and psychology. (4) **Lab-Based Control vs. Remote Scalability**: Traditional lab-based software offers maximum control over the experimental environment, ensuring high internal validity. Emerging web-based platforms promise unprecedented scalability and participant diversity, boosting external validity but introducing significant challenges in maintaining experimental control. Ultimately, these dichotomies, summarized in Fig. 2, demonstrate that the “ideal” software depends entirely on the specific priorities of the research design.

6.2. Identifying gaps and future needs

There is a tool for almost all the niches in the current ecosystem, but there are still several significant gaps. We need a tool that successfully bridges the gap between ease of use and flexibility. It should have the intuitive, GUI-driven design of a tool like NeuroVR or VREX and the underlying programmatic power and extensibility of a framework like UXF. The hybrid model of PTVR, which has a Python front-end and Unity back-end, is a promising step in this direction, but a more fully integrated solution has yet to emerge.

The field would also benefit immensely from greater standardization and interoperability. It is not efficient to do the same process repeatedly, particularly in areas like spatial navigation, and limits the comparability of results across labs. The wider adoption of community standards, like the Lab Streaming Layer (LSL) for data synchronization, is a critical step. Instead of developing monolithic, all-in-one solutions, future development should focus on building more modular, interoperable components that can be shared and used across different frameworks.

The sheer complexity of the software landscape described in this review highlights an important yet often overlooked institutional challenge: the growing need for specialized technical skills within academic research. This is often unrealistic and unsustainable to expect graduate students or postdoctoral fellows to become proficient software engineers who can create and maintain strong, publication-quality experimental software in addition to their scientific training. Research Software

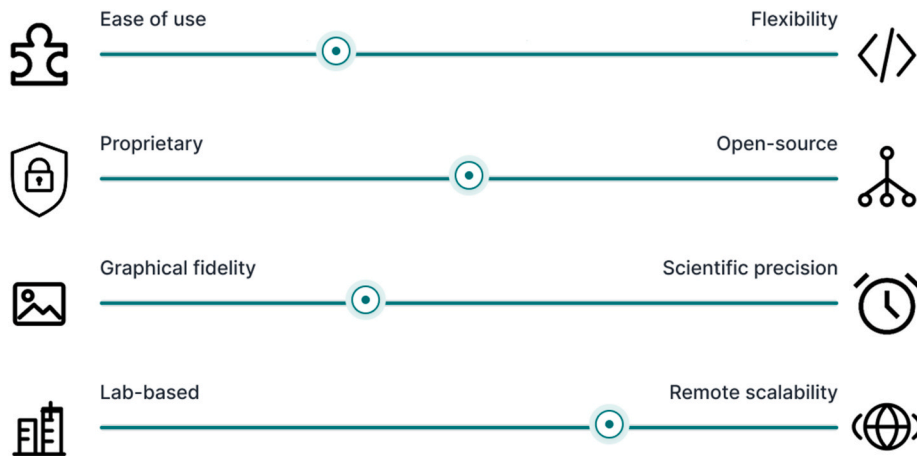


Fig. 2. Critical trade-offs in selecting virtual reality (VR) software for behavioral and cognitive research. Selecting the ideal tool requires balancing four primary dichotomies based on specific research priorities: ease of use vs. experimental flexibility, proprietary support vs. open-source transparency, graphical fidelity vs. timing precision, and lab-based control vs. remote scalability.

Engineers (RSEs) are professional software developers who work in an academic setting. Their work is becoming more and more important to the success of modern, computationally intensive science. Institutions that want to remain at the forefront of fields like VR-based neuroscience need to understand how important this role is and invest in creating career paths and support systems for RSEs as key members of the research enterprise.

6.3. The shift toward open-source and sustainable research ecosystems

While Unity currently remains the de facto standard due to its established ecosystem, the field is beginning to witness a strategic pivot toward open-source and alternative engines. Recent changes in Unity's licensing—particularly the lowering of accessibility for institutions with 'hybrid' funding models—have introduced a level of financial unpredictability that is incompatible with many long-term academic budgets. Consequently, engines like Godot are emerging as high-potential alternatives. As a truly open-source (MIT licensed) tool, Godot offers a 'future-proof' foundation for open science, ensuring that experimental pipelines remain accessible without recurring seat-license costs. Furthermore, the integration of high-fidelity physics in NVIDIA Omniverse and the visual power of Unreal Engine are catering to specific niches in high-end simulation. We anticipate that the next generation of VR toolboxes will likely prioritize platform-agnostic frameworks or open-source engines to ensure the long-term replicability and affordability of behavioral research.

6.4. Final recommendations

The emergence of accessible virtual reality offers a revolutionary opportunity for the behavioral and cognitive sciences, facilitating a paradigm shift towards more ecologically valid and immersive experimental designs. To fully realize this potential, the research community must advance with an emphasis on collaboration, sustainability, and methodological rigor. Funding agencies should consider supporting initiatives that develop and maintain core, open-source community infrastructure to cut down on fragmentation. In turn, researchers should focus on using tools that facilitate transparent and reproducible workflows. Ultimately, the future of this exciting field will depend on more than just the power of our virtual reality hardware. It will also depend on the quality, reliability, and accessibility of the software that brings our scientific questions to life.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used Gemini AI/Google to improve language, flow, and clarity of the text. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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CRediT authorship contribution statement

Mehrnaz Hosseinzadeh: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Morteza Naserbakht:** Conceptualization, Supervision, Writing – review & editing. **Mohammad Fazel Amini:** Data curation, Formal analysis, Investigation, Methodology, Software, Writing – review & editing. **Mohammadmahdi Saeedimehr:** Data curation, Formal analysis, Investigation, Methodology, Writing – original draft, Writing –

review & editing. **Alireza Noori:** Data curation, Formal analysis, Writing – original draft, Writing – review & editing. **Fatemeh Sadat Ghaheiri:** Data curation, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Fatemeh Rafiei:** Data curation, Formal analysis, Writing – original draft, Writing – review & editing. **Elham Abharzad:** Data curation, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Tara Jafar-gholi:** Writing – original draft, Writing – review & editing. **Fatemeh Sadat Mirfazeli:** Conceptualization, Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chbr.2026.101194>.

Data availability

The data supporting this research will be publicly available in the Open Science Framework (OSF) repository after the publication of this article, and can be accessed at <https://doi.org/10.17605/OSF.IO/N6JFC>. Additionally, any further detailed information or raw data will be made available to anyone upon request to the corresponding author.

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